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中国海洋大学全日制本科课程期末考试试卷

2026 年 Spring 季学期　考试科目：Statistical Modelling and Inference

学院：海德学院

试卷类型：SAMPLE EXAM 2　命题人：Thomas Shang　课程教师：Dr. Waqar Hafeez

Examination instructions: This is a closed-book sample examination. A non-programmable calculator is permitted. Show all working. Formulae and statistical quantities required for each question are provided in the question.

Question	1	2	3	4	5	Total
Marks	20	20	20	20	20	100

Name: ____________________ Student ID: ____________________

Major/Year: ____________________ Seat Number: ____________________

Examination Room: ____________________ Instructor: ____________________

Sample Examination 2

Five questions, 20 marks each

End of cover page. Begin Question 1 on the next page.

Question 1 [20 marks total]

Combining two unbiased estimators, bias, variance, and MSE

Suppose $E(\hat{\theta}_1) = E(\hat{\theta}_2) = \theta$, $\mathrm{Var}(\hat{\theta}_1) = 9$, and $\mathrm{Var}(\hat{\theta}_2) = 16$. Define

$$\hat{\theta}_a = a\hat{\theta}_1 + (1-a)\hat{\theta}_2.$$

For an estimator $\hat{\theta}$ of θ,

$$\mathrm{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta, \qquad \mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

For constants a and b,

$$\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y).$$

If the estimators are independent, the covariance term is zero.

(a) Prove that $\hat{\theta}_a$ is unbiased for any fixed a, and state its bias. [4 marks]

(b) Assuming independence, find $\mathrm{Var}(\hat{\theta}_a)$ and $\mathrm{MSE}(\hat{\theta}_a)$. [5 marks]

(c) Find the value of a that minimises the MSE under independence. [5 marks]

(d) Now suppose $\mathrm{Cov}(\hat{\theta}_1, \hat{\theta}_2) = 3$. Find the MSE-minimising value of a. [6 marks]

Question 2 [20 marks total]

Two-sample confidence interval and hypothesis test

A study compares mean weekly study hours for two independent groups of students.

	n	$\bar{y}$	s
Group A	36	18.4	4.8
Group B	49	15.9	5.6

Assume the samples are independent and large enough for a normal approximation.

For large independent samples,

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \qquad Z = \frac{(\bar{Y}_1 - \bar{Y}_2) - \Delta_0}{SE}.$$

Use $z_{0.025} = 1.96$, $z_{0.05} = 1.645$, and $\Phi(2.21) = 0.9864$.

(a) Construct a 95% confidence interval for $\mu_A - \mu_B$. [8 marks]

(b) Test $H_0 : \mu_A - \mu_B = 0$ against $H_a : \mu_A - \mu_B > 0$ at $\alpha = 0.05$. [7 marks]

(c) Compute the p-value using the provided table value. [3 marks]

(d) Interpret the result. [2 marks]

Question 3 [20 marks total]

Multiple linear regression and least squares

Six students are observed. The response y is exam score, x_1 is weekly study hours, and x_2 is number of practice quizzes completed.

i	1	2	3	4	5	6
y_i	54	58	65	70	78	82
x_{1i}	1	2	3	4	5	6
x_{2i}	0	1	1	2	3	3

Fit $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$.

$$(X^\top X)\hat{\beta} = X^\top y, \qquad X^\top X = \begin{pmatrix} 6 & 21 & 10 \\ 21 & 91 & 44 \\ 10 & 44 & 24 \end{pmatrix}, \qquad X^\top y = \begin{pmatrix} 407 \\ 1532 \\ 735 \end{pmatrix}.$$

(a) Write the normal equations. [5 marks]

(b) Solve for $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$. [9 marks]

(c) Give the fitted regression equation. [3 marks]

(d) Predict the score for $x_1 = 5$ and $x_2 = 2$. [3 marks]

Question 4 [20 marks total]

Fitted values and residual properties in simple linear regression

In the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

assume $E(Y_i) = \beta_0 + \beta_1 X_i$, $\mathrm{Var}(Y_i) = \sigma^2$, and independent observations. Let

$$\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 X_0, \qquad \hat{e}_i = Y_i - \hat{Y}_i.$$

You may use

$$E(\hat{\beta}_0) = \beta_0, \quad E(\hat{\beta}_1) = \beta_1, \quad \mathrm{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{S_{xx}} \right),$$

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \qquad \mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{X}}{S_{xx}}.$$

(a) Prove $E(\hat{Y}_0) = \beta_0 + \beta_1 X_0$. [5 marks]

(b) Derive $\mathrm{Var}(\hat{Y}_0) = \sigma^2 \left(1/n + (X_0 - \bar{X})^2 / S_{xx}\right)$. [9 marks]

(c) Prove $\sum_{i=1}^{n} \hat{e}_i = 0$. [4 marks]

(d) State why fitted-value variance is smallest at $X_0 = \bar{X}$. [2 marks]

Question 5 [20 marks total]

Normal MLE and Fisher information

Let $Y_1, \ldots, Y_n \overset{iid}{\sim} N(\mu, \sigma^2)$.

The normal density is

$$f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y-\mu)^2}{2\sigma^2}\right\}.$$

For parameter vector (μ, v) with $v = \sigma^2$,

$$I(\mu, v) = -E\{H(\mu, v)\},$$

where H is the Hessian matrix of the log-likelihood.

(a) Assume $v = \sigma^2$ is known. Derive the MLE of μ. [5 marks]

(b) Now assume both μ and $v = \sigma^2$ are unknown. Write $\ell(\mu, v)$. [4 marks]

(c) Derive the score components $\partial\ell/\partial\mu$ and $\partial\ell/\partial v$. [5 marks]

(d) State the Fisher information matrix for (μ, v). [4 marks]

(e) Explain why the cross-information term is zero. [2 marks]

End of examination questions.

Check that you have attempted all five questions.